

Ubuntu 16.04.6 LTS mininet-vm tty1

mininet-vm login: mininet

Password:

Last login: Sun Dec 21 00:12:40 PST 2025 from 192.168.159.130 on pts/0

Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

- * Documentation: <https://help.ubuntu.com>
- * Management: <https://landscape.canonical.com>
- * Support: <https://ubuntu.com/advantage>

mininet@mininet-vm:~\$ ip a

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:75:b8:7e brd ff:ff:ff:ff:ff:ff
    inet 192.168.159.136/24 brd 192.168.159.255 scope global eth0
        valid_lft forever preferred_lft forever
3: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1
    link/ether a6:2a:be:14:2e:a2 brd ff:ff:ff:ff:ff:ff
4: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1
    link/ether 26:c7:69:c2:fd:46 brd ff:ff:ff:ff:ff:ff
mininet@mininet-vm:~$
```

(kalipodo_saha@Kalipodo)-[~]
\$ ssh mininet@192.168.159.136

** WARNING: connection is not using a post-quantum key exchange algorithm.

** This session may be vulnerable to "store now, decrypt later" attacks.

** The server may need to be upgraded. See <https://openssh.com/pq.html>

mininet@192.168.159.136's password:

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New release '18.04.6 LTS' available.

Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Jan 5 11:35:46 2026

mininet@mininet-vm:~\$ |

mininet@mininet-vm:~\$ ifconfig

```
eth0      Link encap:Ethernet  HWaddr 00:0c:29:75:b8:7e
          inet addr:192.168.159.136  Bcast:192.168.159.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:425 errors:0 dropped:0 overruns:0 frame:0
          TX packets:318 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:39600 (39.6 KB)  TX bytes:31160 (31.1 KB)
```

```
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1082 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1082 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:54100 (54.1 KB)  TX bytes:54100 (54.1 KB)
```

```
mininet@mininet-vm:~$ sudo mn --topo single,3
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> |
```

```
mininet> nodes
available nodes are:
c0 h1 h2 h3 s1
```

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
c0
```

```
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1891>
<Host h2: h2-eth0:10.0.0.2 pid=1894>
<Host h3: h3-eth0:10.0.0.3 pid=1897>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=1903>
<Controller c0: 127.0.0.1:6653 pid=1884>
mininet> h1 ifconfig -a
h1-eth0 Link encap:Ethernet HWaddr 52:04:06:1e:66:f4
        inet addr:10.0.0.1 Bcast:10.255.255.255 Mask:255.0.0.0
        UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

lo Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        UP LOOPBACK RUNNING MTU:65536 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1
        RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

```
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.925 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.165 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.040 ms

— 10.0.0.2 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.038/0.241/0.925/0.345 ms
mininet> |
```

ifconfig - help gives usage information.

```
mininet> s1 ifconfig -a
eth0      Link encap:Ethernet  HWaddr 00:0c:29:75:b8:7e
          inet addr:192.168.159.136  Bcast:192.168.159.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1005 errors:0 dropped:0 overruns:0 frame:0
          TX packets:679 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:85751 (85.7 KB)  TX bytes:67392 (67.3 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1798 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1798 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:92580 (92.5 KB)  TX bytes:92580 (92.5 KB)

ovs-system Link encap:Ethernet  HWaddr 92:d1:69:1f:05:3f
          BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s1        Link encap:Ethernet  HWaddr 26:c7:69:c2:fd:46
          BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
s1-eth1   Link encap:Ethernet  HWaddr c6:9c:4f:bf:6f:de
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s1-eth2   Link encap:Ethernet  HWaddr 9e:99:ea:77:5d:11
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s1-eth3   Link encap:Ethernet  HWaddr 66:e8:ca:b2:9d:2b
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```